

**Project Design Phase Solution
Architecture**

Date	04 July 2026
Project Name	Cosmetic Insights – Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behaviour, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

KEY ASPECTS OF YOUR SOLUTION ARCHITECTURE

1. User Layer (Entry Point)

o Users include:

Cosmetic consumers: use filters to discover products

Marketers: analyse trends and user behaviour

o Interface should be simple, responsive web

interface with embedded Tableau dashboard

2. User Interface (Frontend) o Technologies: HTML, CSS, Bootstrap, JavaScript

- o Features:

- Filters: skin type, budget, brand, rating
 - Siddh and comparison interfere
 - Embedded Tableau dashboards for date visuals

3. Application Layer (Backend)

- o Technologies: Python
- (Folds/Streamlet) or JavaScript (Node.js)
- o Functions:
 - Pass user inputs (filters) to Tableau
 - Handle login/duty (if used)
 - Process and fetch date from review/product databases
- Optional:
 - route feedback submission

4. Visualization Layer (Dashboard)

- o Tool: Tableau Public or Tableau Server
- o Displays:
 - Top trending cosmetic products
 - Charts by skin type, rating, region, additory
 - Comparison visuals for pride, review score, ingredients

5. Data Sources

- o Sources:
 - Product : CSV / MySQL / Google Sheets
 - Review : Customer ratings & feedback
- o Features: Structured datasets allow filtering and hurting Can be updated regularly or live connected

6. Feedback Engine (Optional)

- o Collects product ratings/reviews from users
- o Stores in review repository
- o Feeds into Tableau to update user satisfaction visuals

7. Hosting&Storage

- o Tableau Public: for hosting dashboards
- o GitHub Pages / Heroku: for web door frontend
- o Google Drive / Firebase: for storing files and datasets

8. Data Flow

- o User → UI → Addend → Tabled → Visualization
- o Optionally, Feedback → Stored → Updates Dashboard

9. Scalability

- o Add more products detests
- o Support multi-Theory expansion: skidder, make-up, fragrance
- o Add AI/ML trend

10. Security (Optional/Advanced)

- Auth for secure login (Google, LinkedIn)
- backend validation for did inputs and review Authenticity

Example - Solution Architecture Diagram:

Cosmetic Insights:

Navigating Cosmetics Trends and Consumer Insights with Tableau

